Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	I II	Ultrafiltration – glomerulus / Bowman's capsule (1) Reabsorption – proximal convoluted tubule (1)	2			2		
		(ii)		X-{ascending limb/ loop of Henle} **and** Y-collecting duct (1) Osmoregulation (1)	2			2		
	(b)	(i)		(Conc of urea increases because) {little/no} urea is reabsorbed but water is reabsorbed (1) (Conc of chloride decreases because) Chloride is {reabsorbed /diffuses out /actively transported out}(faster than water) (1)			2	2		2
		(ii)		A. (Concentration of glucose decreases because) glucose is reabsorbed (faster than water) (1) **Any three (x1) from** B. With inhibitor no respiration and therefore {no/less} ATP (1) C. No active transport of Na^+ (out of PCT) (1) D. Therefore no concentration gradient of Na^+ / owtte E. For the co-transport of glucose (1)			4	4		
	(c)	(i)		{To compare concentration/ show the loss} (of Na^+) {in loop of Henle/ in descending and ascending limbs/ between PCT and DCT/ before and after absorption}			1	1		1

Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
		(ii)		**Any four (x1) from:** A. Descending limb water leaves by osmosis (1) B. Na^+ concentration increases {as it moves down the descending limb/ as it get closer to the {hairpin/ apex}/ as it moves deeper into the medulla} (1) C. Ascending limb Na^+ pumped out (1) D. Ascending limb is impermeable to water (1) E. Na^+ concentration decreases as it moves up the ascending limb (1)	4			4		
				Question 5 total	**8**	**0**	**7**	**15**	**0**	**3**